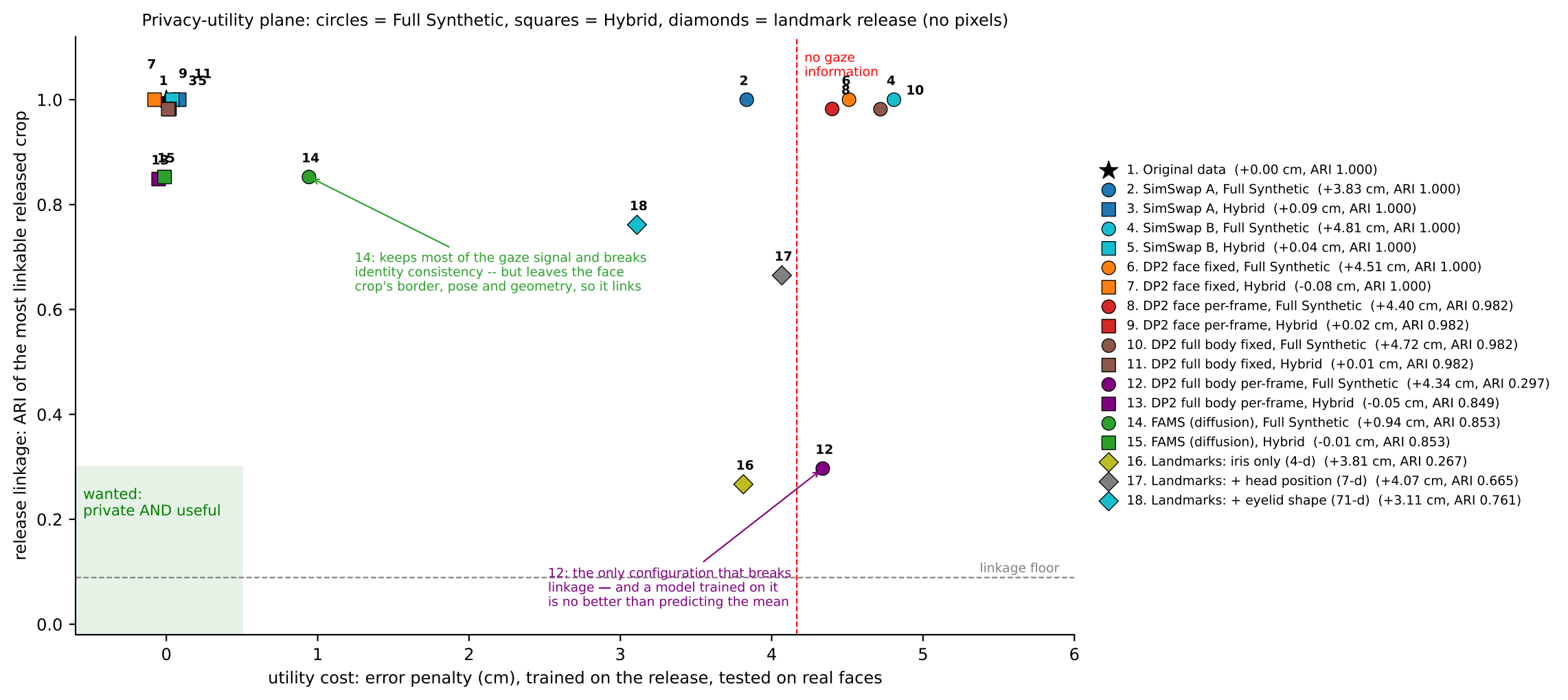
Privacy-utility plane: circles = Full Synthetic, squares = Hybrid, diamonds = landmark release (no pixels)
release linkage: ARI of the most linkable released crop
utility cost: error penalty (cm), trained on the release, tested on real faces
no gaze information
linkage floor
wanted:
private AND useful
14: keeps most of the gaze signal and breaks identity consistency -- but leaves the face crop's border, pose and geometry, so it links
12: the only configuration that breaks linkage — and a model trained on it is no better than predicting the mean
1. Original data (+0.00 cm, ARI 1.000)
2. SimSwap A, Full Synthetic (+3.83 cm, ARI 1.000)
3. SimSwap A, Hybrid (+0.09 cm, ARI 1.000)
4. SimSwap B, Full Synthetic (+4.81 cm, ARI 1.000)
5. SimSwap B, Hybrid (+0.04 cm, ARI 1.000)
6. DP2 face fixed, Full Synthetic (+4.51 cm, ARI 1.000)
7. DP2 face fixed, Hybrid (-0.08 cm, ARI 1.000)
8. DP2 face per-frame, Full Synthetic (+4.40 cm, ARI 0.982)
9. DP2 face per-frame, Hybrid (+0.02 cm, ARI 0.982)
10. DP2 full body fixed, Full Synthetic (+4.72 cm, ARI 0.982)
11. DP2 full body fixed, Hybrid (+0.01 cm, ARI 0.982)
12. DP2 full body per-frame, Full Synthetic (+4.34 cm, ARI 0.297)
13. DP2 full body per-frame, Hybrid (-0.05 cm, ARI 0.849)
14. FAMS (diffusion), Full Synthetic (+0.94 cm, ARI 0.853)
15. FAMS (diffusion), Hybrid (-0.01 cm, ARI 0.853)
16. Landmarks: iris only (4-d) (+3.81 cm, ARI 0.267)
17. Landmarks: + head position (7-d) (+4.07 cm, ARI 0.665)
18. Landmarks: + eyelid shape (71-d) (+3.11 cm, ARI 0.761)